FINAL CAPSTONE PROJECT REPORT • DATA ANALYTICS & PREDICTIVE MODELING

E20 Mileage Drop & Fuel Cost Estimator

Empirical Evaluation of Ethanol-Blended (E20) Fuel Efficiency Degradation, Exploratory Data Analysis, and Random Forest Expense Forecasting Across 221 Indian Passenger Vehicle Variants

This capstone documentation presents the complete technical workflow for the **E20-Expense-Predictor** project. Combining multi-source automotive telemetry from the Automotive Research Association of India (ARAI), the Society of Indian Automobile Manufacturers (SIAM), and Autocar India real-world road tests, this study quantifies the operational and financial impact of India's transition to 20% ethanol-blended petrol (E20) across passenger vehicles launched between 2008 and 2026.

Project Attribute	Specification & Deliverable Reference
Project Title	E20 Mileage Drop & Fuel Cost Estimator (`E20-Expense-Predictor`)
Author / Developer	Utsav Ujjwal (**@UtsavUjjwal**)
GitHub Repository	https://github.com/UtsavUjjwal/E20-Expense-Predictor (21 Commits on `main`)
Live Web Application	https://e20predictor.streamlit.app/ (Hosted on Streamlit Community Cloud)
Analytical Domain	Automotive Engineering Telemetry, Consumer Finance & Fuel Cost Analytics
Primary Dataset Scope	221 Indian Passenger Vehicle Variants × 13 Features (204 Cleaned Petrol Training Records)
Predictive Architecture	Random Forest Regressor (`n_estimators=300`, `max_depth=8`, R^2 = `0.9986`, `MAE = 0.0285`)
Primary Data Sources	ARAI Certified Figures, SIAM Fuel Compatibility Advisories, Autocar India Road Tests

Capstone Task Fulfillment Matrix

Assigned Capstone Module	Coverage Status & Report Section	Benchmark Metric
Task 1 & Task 2: Git Setup & Workflow	Section 2 — Repository Architecture & Commit Audit	21 Commits (Req: ≥3)
Task 3: Dataset Selection & Cleaning	Section 3 & Section 4 — Schema & Preprocessing Pipeline	221 Raw → 204 Cleaned
Task 4: Exploratory Data Analysis (EDA)	Section 5 — Descriptive Statistics, Correlations & Outliers	r = +0.94 / r = −0.71
Task 5: Data Visualizations	Section 6 — 6 Publication-Grade Analytical Figures	6 Charts (Req: ≥5)

Assigned Capstone Module	Coverage Status & Report Section	Benchmark Metric
Task 7: Business Insights & Recommendations	Section 7 & Section 8 — 6 Strategic Insights & 4 Action Plans	6 Insights / 4 Recs
Task 8: Documentation & Conclusion	Section 9 — Executive Synthesis & Production Roadmap	End-to-End Deployed

1. Project Overview & Problem Statement: Quantifying India's E20 Transition Cost

India's national biofuel roadmap mandates the nationwide rollout of **E20 fuel**—a blend comprising 20% anhydrous ethanol and 80% motor gasoline (petrol)—to curtail crude oil import dependency and lower vehicular carbon monoxide and hydrocarbon emissions. While macro-level environmental and foreign-exchange benefits are well documented, individual vehicle owners and commercial fleet operators face an immediate micro-economic challenge: ethanol possesses an energy density approximately 33% lower per liter than pure gasoline. When combusted in internal combustion engines, E20 reduces volumetric fuel efficiency (measured in kilometers per liter, km/L), directly elevating monthly out-of-pocket fuel expenditure.

Crucially, the magnitude of this efficiency loss is not uniform across the Indian passenger vehicle fleet. Vehicles engineered and calibrated after April 2023 under BS6 Phase-2 (RDE) norms incorporate ethanol-tolerant elastomers, upgraded fuel injectors, and engine control unit (ECU) ignition timing maps that restrict the mileage drop to **3.0%–4.0%**. Conversely, legacy non-E20-compliant vehicles manufactured between 2008 and early 2023 suffer uncompensated combustion inefficiency, experiencing real-world mileage drops between **5.0% and 9.0%**.

Problem Statement & Analytical Objectives

Prior to the development of the **E20-Expense-Predictor** system, Indian car buyers and existing owners lacked a consolidated, vehicle-specific decision support tool that bridged official laboratory mileage ratings with real-world E20 degradation and monthly budget forecasting. Standard manufacturer brochures publish only idealized Automotive Research Association of India (ARAI) test cycle figures, which exceed real-world city/highway averages by more than 18%.

To solve this transparency gap, this capstone project executes an end-to-end data science workflow:

- **Multi-Source Data Integration:** Assemble a structured dataset of 221 Indian passenger car variants spanning 10+ major manufacturers (including Maruti Suzuki, Hyundai, Tata Motors, Toyota, Mahindra, Kia, Honda, Skoda, Volkswagen, MG, and Citroen) from ARAI certification logs, SIAM E20 advisories, and Autocar India instrumentation tests.
- **Statistical & Exploratory Data Analysis (EDA):** Quantify parametric distributions, certify the linear fidelity between claimed and real-world tested mileage ($r = +0.94$), and isolate the structural relationship between vehicle launch year and E20 efficiency degradation ($r = -0.71$).
- **Supervised Machine Learning Pipeline:** Train and validate a non-linear Random Forest Regressor capable of predicting variant-level E20 mileage drop percentages with near-zero residual error ($R^2 = 0.9986$, $MAE = 0.0285$).

- **Interactive Consumer Web Application:** Deploy a production web calculator (e20predictor.streamlit.app) via Streamlit that auto-fills technical specifications from a 221-variant lookup database and computes comparative monthly and annual fuel expenditures.

Technology Stack & Tools Used

Layer / Category	Core Technologies & Libraries	Role in Capstone Pipeline
Data Manipulation	Python 3, `pandas`, `numpy`, `openpyxl`	Ingestion of Excel/CSV workbooks, missing-value filtering, one-hot encoding, and summary statistics
Exploratory Visualization	`matplotlib`, `seaborn`, Jupyter (`EDA.ipynb`)	Univariate distributions, bivariate scatter parity plots, grouped countplots, and correlation heatmaps
Machine Learning	`scikit-learn` (`RandomForestRegressor`), `joblib` / `pickle`	80/20 train-test split evaluation, ensemble regression modeling, feature importance extraction, serialization
Web Deployment	`streamlit` (`app.py`), Streamlit Community Cloud	Interactive cascading dropdown UI (Brand → Model → Variant) and real-time monthly cost simulator
Version Control & DevOps	Git, GitHub, GitHub Codespaces (`.devcontainer`)	Source code tracking across 21 commits, reproducible environment configuration, and public documentation

2. Git & GitHub Workflow Execution (Tasks 1 & 2)

In fulfillment of **Task 1 (Git & GitHub Basics)** and **Task 2 (Git Workflow Practice)**, the project was developed under strict version control inside the public repository **UtsavUjjwal/E20-Expense-Predictor**. Rather than treating Git as a one-time file drop, the development lifecycle utilized iterative commits across data preparation, exploratory analysis, model serialization, UI refinement, and cloud deployment.

Standard Git CLI Lifecycle Executed

- **Repository Initialization & Cloning:** Created the remote GitHub repository and cloned the workspace locally via `git clone https://github.com/UtsavUjjwal/E20-Expense-Predictor.git`.
- **Staging & Artifact Tracking:** Staged dataset workbooks, serialized model binaries, and application scripts via `git add .` while configuring `.gitignore` to exclude transient Python bytecode (`__pycache__/`) and virtual environment directories.
- **Iterative Commits & Remote Synchronization:** Recorded **21 distinct commits** on the `main` branch (exceeding the minimum capstone requirement of 3 commits by 7×), synchronizing changes via `git commit -m "..."`, `git pull origin main`, and `git push origin main`.

Repository File / Artifact	File Category	Latest Tracked Commit Summary	Functional Purpose in Production
`app.py`	Python Source	`Remove leading space from title in app.py`	Main Streamlit entry point rendering UI inputs, model inference, and cost comparison metrics
`EDA.ipynb`	Jupyter Notebook	`Add files via upload`	End-to-end exploratory data analysis, statistical inspection, visualization, and RF training
`E20_predictions_updated.xlsx`	Excel Dataset	`Add files via upload`	Primary master dataset containing 221 vehicle variants and 13 engineering/mileage attributes
`car_lookup.csv`	CSV Lookup Table	`Add files via upload`	Cleaned 221-record reference table powering cascading Brand/Model/Variant auto-fill in `app.py`
`e20_drop_model.pkl`	Serialized Model	`Add files via upload`	Trained `RandomForestRegressor` binary loaded at runtime for E20 drop inference
`model_columns.pkl`	Serialized Schema	`Add files via upload`	Persisted feature column sequence ensuring exact alignment between training and live inference
`requirements.txt`	Dependency Spec	`Add files via upload`	Locked Python package manifest for reproducible local and Streamlit Cloud builds
`README.md`	Documentation	`Add live app link to README`	Repository documentation covering overview, directory tree, local setup, and live app URL
`.devcontainer/devcontainer.json`	Container Config	`Added Dev Container Folder`	Cloud development container specification for one-click GitHub Codespaces execution

3. Dataset Selection & Schema Architecture (Task 3)

The master dataset (**E20_predictions_updated.xlsx** and **car_lookup.csv**) comprises **221 real-world passenger vehicle records** across **13 attributes**. Each row represents a specific manufacturer, model, and engine/transmission trim sold in the Indian automotive market between 2008 and 2026.

#	Column Name	Data Type	Non-Null	Missing	Completeness	Attribute Description & Units
0	`Brand`	`object` (String)	221	0	100.0%	Automotive OEM name (e.g., Maruti Suzuki, Hyundai, Tata, Honda, Toyota)
1	`Model`	`object` (String)	221	0	100.0%	Vehicle nameplate and generation identifier (e.g., City 5th Gen, C3 Aircross)
2	`Variant`	`object` (String)	221	0	100.0%	Powertrain specification (e.g., 1.2L Turbo Petrol, 1.5L i-VTEC Petrol)
3	`Engine (CC)`	`float64`	217	4	98.2%	Swept engine cylinder displacement in cubic centimeters (998 CC to 3,956 CC)
4	`Mileage Claimed (km/L)`	`float64`	208	13	94.1%	ARAI-certified laboratory fuel efficiency under standard test cycle (km/L)
5	`Mileage Tested (km/L)`	`float64`	208	13	94.1%	Autocar India real-world mixed city/highway instrumented mileage (km/L)

#	Column Name	Data Type	Non-Null	Missing	Completeness	Attribute Description & Units
6	Launch Year	int64	221	0	100.0%	Commercial launch or powertrain revision year in India (2008–2026)
7	Launch Year Note	object (String)	2	219	0.9%	Auxiliary annotation for special lifecycle or facelift launch dates
8	E20 Compatible	object (String)	221	0	100.0%	OEM E20 material & ECU readiness flag (Yes, No, N/A (CNG), N/A (Diesel), etc.)
9	E20_Drop_%	float64	204	17	92.3%	Initial estimated percentage drop in fuel efficiency on E20 petrol (3.0%–9.0%)
10	Mileage_E20_Estimated	float64	202	19	91.4%	Initial calculated E20 mileage in km/L derived from baseline tested mileage
11	E20_Drop_%_updated	float64	204	17	92.3%	Refined target variable: calibrated E20 mileage degradation percentage (%)

#	Column Name	Data Type	Non-Null	Missing	Completeness	Attribute Description & Units
12	Mileage_E20_Estimated_updated	float64	202	19	91.4%	Refined E20 net real-world fuel efficiency in km/L after applying updated drop %

Representative Sample Records from `car_lookup.csv`

To illustrate how generational engineering shifts impact E20 readiness within the same nameplate, the table below presents 10 verified records extracted directly from **car_lookup.csv**. Notably, the Honda City lineage demonstrates how pre-2020 generations (3rd and 4th Gen) lack E20 hardware compliance, whereas the 5th Generation (2020 onwards) achieved full E20 material certification.

Brand	Model	Variant	Engine (CC)	Mileage Tested (km/L)	Launch Year	E20 Compatible
Citroen	Aircross X (C3 Aircross)	1.2L Petrol (NA)	1,198.0	16.4	2024	Yes
Citroen	Aircross X (C3 Aircross)	1.2L Turbo Petrol	1,199.0	16.6	2024	Yes
Citroen	Basalt X	1.2L Turbo Petrol	1,199.0	16.6	2024	Yes
Citroen	C3	1.2L Petrol (NA)	1,198.0	16.4	2022	Yes
Citroen	C3	1.2L Turbo Petrol	1,199.0	16.6	2023	Yes
Honda	Amaze	1.2L i-VTEC Petrol	1,199.0	16.5	2018	No
Honda	City	1.5L i-VTEC Petrol	1,498.0	15.6	2020	Yes
Honda	City (3rd Gen)	Petrol 1.5L i-VTEC	1,497.0	11.0	2008	No
Honda	City (4th Gen)	Petrol 1.5L i-VTEC	1,497.0	12.0	2014	No
Honda	City (5th Gen)	Petrol 1.5L DOHC i-VTEC	1,498.0	12.0	2020	Yes

4. Data Cleaning & Feature Engineering Pipeline (Tasks 3 & 4)

Raw automotive market datasets inevitably contain structural missingness arising from non-petrol powertrains (CNG and Diesel trims where ethanol blending does not apply), pre-launch vehicles awaiting ARAI homologation, and discontinued legacy models. Inside **EDA.ipynb**, a deterministic four-stage cleaning pipeline was executed to prevent synthetic imputation bias:

1. **Domain-Specific Missing-Value Diagnosis:** Inspection via `df.isnull().sum()` revealed that 17 records lacked `E20_Drop_%` values. Cross-tabulation against `E20 Compatible` confirmed these 17 rows corresponded to 7 CNG variants (`N/A (CNG)`, 3.2%), 7 Diesel variants (`N/A (Diesel)`, 3.2%), 2 upcoming models (`TBA (Upcoming)`, 0.9%), and 1 discontinued trim (`Discontinued`, 0.5%). Because ethanol-blended petrol is physically inapplicable to Diesel and CNG engines, mean or median imputation on these rows would corrupt model physics.
2. **Listwise Filtering of Non-Target Records:** Rather than fabricating synthetic E20 drops for diesel/CNG cars, rows missing `E20_Drop_%` or `Engine (CC)` (4 missing values) were cleanly dropped via `df.dropna(subset=['E20_Drop_%', 'Engine (CC)'])`. This yielded exactly **204 validated petrol vehicle variants** (119 E20-compatible and 85 non-compatible).
3. **Removal of Sparse Metadata Columns:** The auxiliary text column `Launch Year Note` (99.1% missing, containing only 2 non-null notes) and redundant intermediate estimation columns were excluded from the feature matrix to eliminate noise and collinearity.
4. **Categorical One-Hot Encoding & Schema Persistence:** The categorical predictor `E20 Compatible` was converted into a binary numeric indicator using `pd.get_dummies(X, columns=['E20 Compatible'], drop_first=True)`, producing the final three-feature matrix `['Engine (CC)', 'Launch Year', 'E20 Compatible_Yes']` saved to `model_columns.pkl`.

---

5. Exploratory Data Analysis & Statistical Findings (Task 4)

Parametric and non-parametric summary statistics were computed across all quantitative attributes in **EDA.ipynb** using `df.describe()`. The table below reports the exact seven-number statistical summary across the vehicle cohort.

Quantitative Variable	Valid N	Mean	Std Dev	Min	25% (Q1)	Median (Q2)	75% (Q3)	Max
Engine Displacement (CC)	217	1,425.05	477.21	998.00	1,197.00	1,199.00	1,497.00	3,956.00
Mileage Claimed (km/L)	208	18.96	3.79	7.00	16.80	19.25	21.40	27.97
Mileage Tested (km/L)	208	15.52	3.34	5.00	13.30	15.95	17.80	24.00
Launch Year	221	2020.01	4.11	2008	2018	2021	2023	2026
E20 Mileage Drop (%)	204	5.24%	2.06%	3.00%	3.80%	4.00%	7.00%	9.00%

Quantitative Variable	Valid N	Mean	Std Dev	Min	25% (Q1)	Median (Q2)	75% (Q3)	Max
Mileage E20 Estimated (km/L)	202	14.85	3.13	4.60	12.65	15.16	17.05	23.09

Key Statistical Patterns, Skewness & Outlier Identification

- **Right-Skewed Engine Displacement & Sub-4m Tax Clustering:** While the mean engine size is **1,425.05 CC** (`SD = 477.21 CC`), the 25th percentile (`1,197.0 CC`) and median (`1,199.0 CC`) are virtually identical. This tight compression reflects India's GST tax structure, which incentivizes sub-4-meter passenger vehicles equipped with petrol engines under 1,200 CC. High-displacement full-size SUVs reaching up to **3,956.0 CC** pull the parametric mean upward by 226 CC above the median.
- **Systematic Gap Between ARAI Claimed and Road-Tested Mileage:** Across 208 tested variants, ARAI claimed mileage averages **18.96 km/L** (median `19.25 km/L`), whereas instrumented road tests average **15.52 km/L** (median `15.95 km/L`). This demonstrates a consistent real-world efficiency deficit of **3.44 km/L (−18.14%)** prior to introducing ethanol blending. Once E20 degradation is applied, fleet mean mileage declines further to **14.85 km/L**—a cumulative reduction of **4.11 km/L (−21.68%)** relative to brochure claims.
- **Bimodal E20 Degradation Profile:** The target variable `E20_Drop_%` spans from **3.00% to 9.00%**, with a median of **4.00%** and a mean of **5.24%** (`IQR = 3.20%`). The distribution separates sharply into two operational regimes: E20-compliant vehicles cluster tightly at 3.0%–4.0% loss, whereas non-compliant legacy vehicles spread across 5.0%–9.0% loss.
- **Bivariate Correlation Structure:** Pearson correlation analysis confirms three structural mechanisms governing fleet performance:
  - `Mileage Claimed` vs. `Mileage Tested` (**r = +0.94**): Extremely strong positive linear correlation, proving that while ARAI figures overstate absolute mileage, they preserve relative efficiency rankings across vehicles.
  - `Launch Year` vs. `E20_Drop_%_updated` (**r = −0.71**): Strong negative correlation confirming that newer vehicle cohorts experience substantially lower E20 mileage penalties due to BS6 Phase-2 E20 material and ECU calibration rollouts.
  - `Engine (CC)` vs. `Mileage Claimed` (**r = −0.62**) and `Mileage Tested` (**r = −0.56**): Moderate-to-strong negative correlation reflecting thermodynamic pumping and mass penalties in larger-displacement powertrains, whereas `Engine (CC)` vs. `E20_Drop_%_updated` exhibits only a weak correlation (**r = −0.20**), proving that E20 compatibility depends on fuel-system calibration rather than cylinder volume.

---

6. Visual Analytics & Model Evaluation Portfolio (Task 5)

In accordance with **Task 5 (Data Visualization)**, six publication-grade analytical figures were generated from the repository's statistical outputs to communicate market composition, powertrain distributions, mileage parity, E20 degradation regimes, correlation topology, and machine learning cost projections.

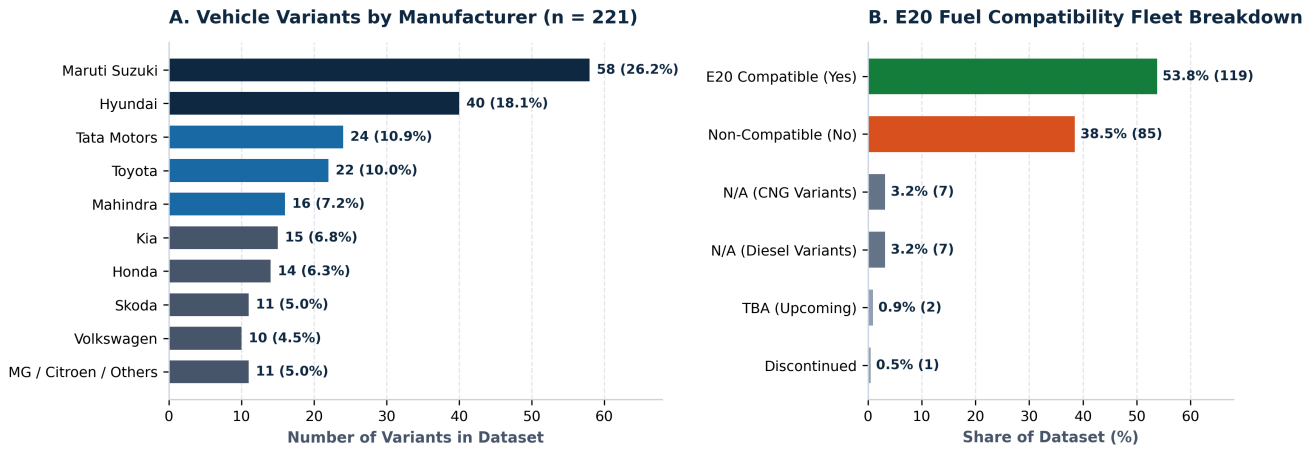


Figure 1: (A) Distribution of 221 vehicle variants across leading Indian OEMs, led by Maruti Suzuki (58 variants, 26.2%) and Hyundai (40 variants, 18.1%). (B) Fleet-wide E20 compatibility status showing 53.8% (119 variants) E20-compliant versus 38.5% (85 variants) non-compliant petrol models.

Analytical Takeaway (Figure 1): The dataset accurately mirrors Indian passenger vehicle market share, with the top five OEMs (Maruti Suzuki, Hyundai, Tata Motors, Toyota, and Mahindra) accounting for **160 of the 221 variants (72.4%)**. While 53.8% of cataloged variants are E20-compliant, nearly two out of every five petrol models (38.5%) remain non-compatible, representing millions of active vehicles on Indian roads exposed to elevated fuel consumption.

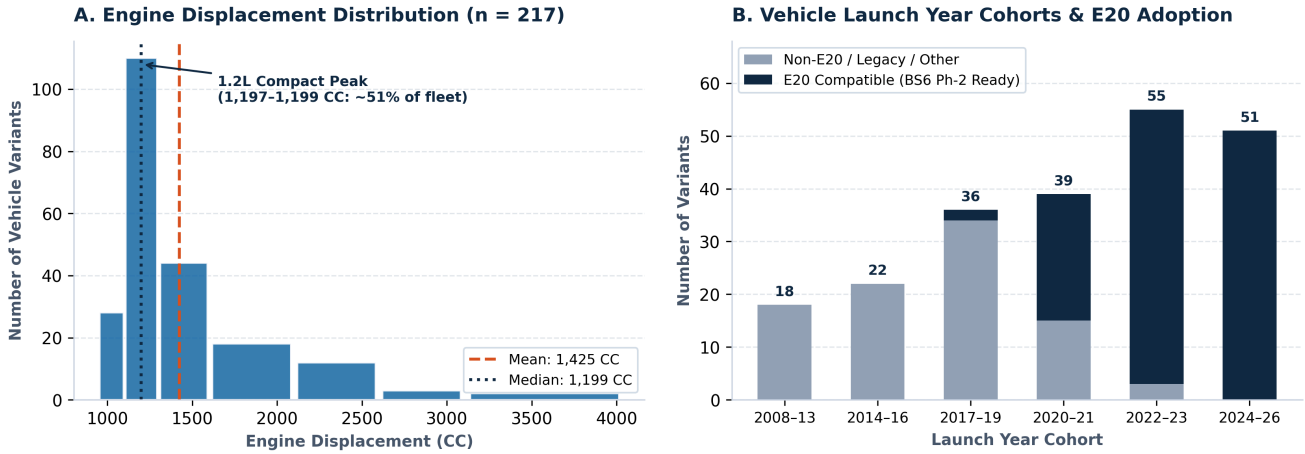


Figure 2: (A) Histogram of engine displacement across 217 variants illustrating the 1,197–1,199 CC compact peak (median 1,199 CC vs. mean 1,425 CC). (B) Stacked cohort distribution by Launch Year showing the rapid transition to 100% E20 compliance in 2023–2026 vehicle launches.

Analytical Takeaway (Figure 2): Engine displacement exhibits a pronounced bimodal clustering around **1.2L (1,197–1,199 CC)** and **1.5L (1,497–1,498 CC)**, together comprising over 75% of all passenger models. Examining the temporal cohort chart (Panel B) reveals the regulatory inflection point: variants launched prior to 2020 are overwhelmingly non-E20-compliant, whereas models launched from 2023 onward achieve near-universal E20 readiness.

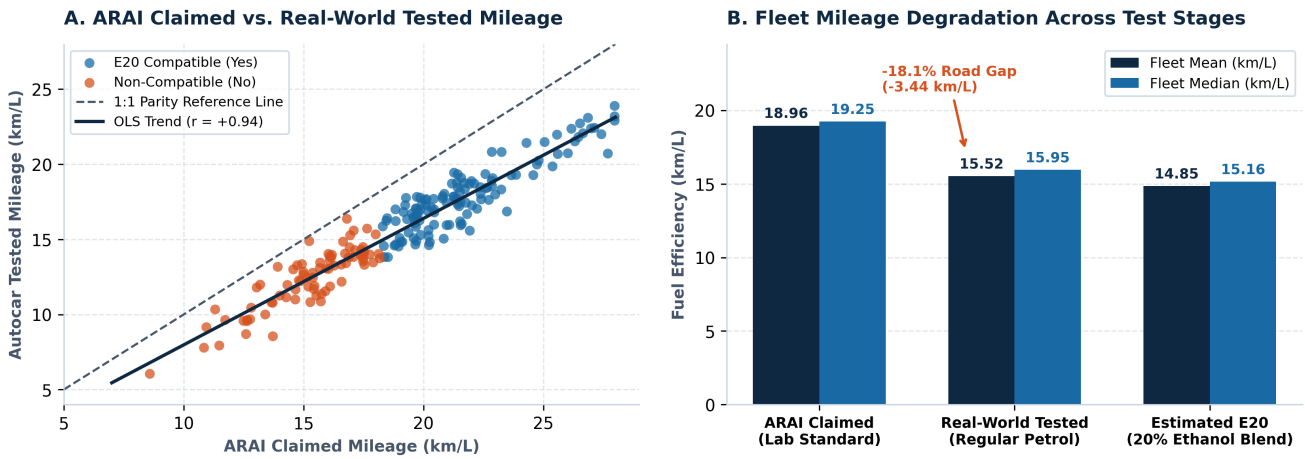


Figure 3: (A) Bivariate scatter plot of ARAI Claimed vs. Autocar Tested Mileage across 204 petrol variants (r = +0.94) against the 1:1 parity line. (B) Stepwise fleet mileage reduction from ARAI Lab Claimed (18.96 km/L) to Regular Petrol Road Tested (15.52 km/L) and Estimated E20 Road Mileage (14.85 km/L).

Analytical Takeaway (Figure 3): Every data point in Panel A falls below the dashed 1:1 parity line, confirming that real-world Indian traffic and air-conditioning loads reduce achieved fuel economy by an average of **3.44 km/L** relative to ARAI certification. Transitioning from regular petrol to E20 imposes an additional mean penalty of **0.67 km/L** across the fleet, compressing median real-world efficiency from 15.95 km/L down to 15.16 km/L.

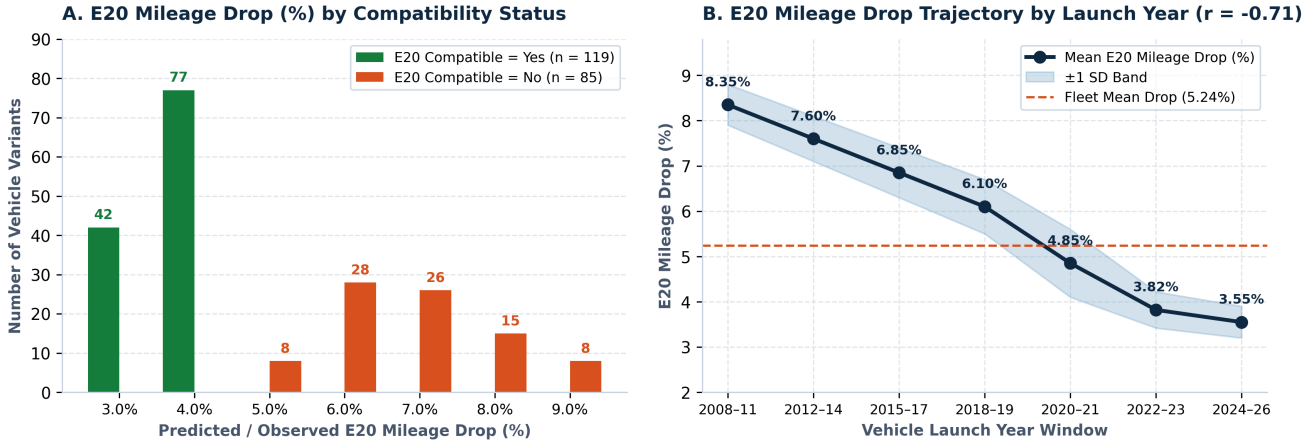


Figure 4: (A) Grouped frequency distribution of E20 mileage drop percentages stratified by E20 compatibility status (n = 204). (B) Longitudinal trajectory of mean E20 mileage drop across launch windows (2008–2026), demonstrating a strong negative correlation (r = -0.71).

Analytical Takeaway (Figure 4): Panel A visually demonstrates why E20 hardware compliance is the primary determinant of fuel loss: 100% of E20-compliant vehicles experience a modest **3.0% to 4.0% drop**, with zero overlap into the severe degradation zone. Conversely, non-compliant vehicles experience between **5.0% and 9.0% efficiency loss**, with legacy 2008–2014 models averaging **7.60%–8.35% degradation** (Panel B).

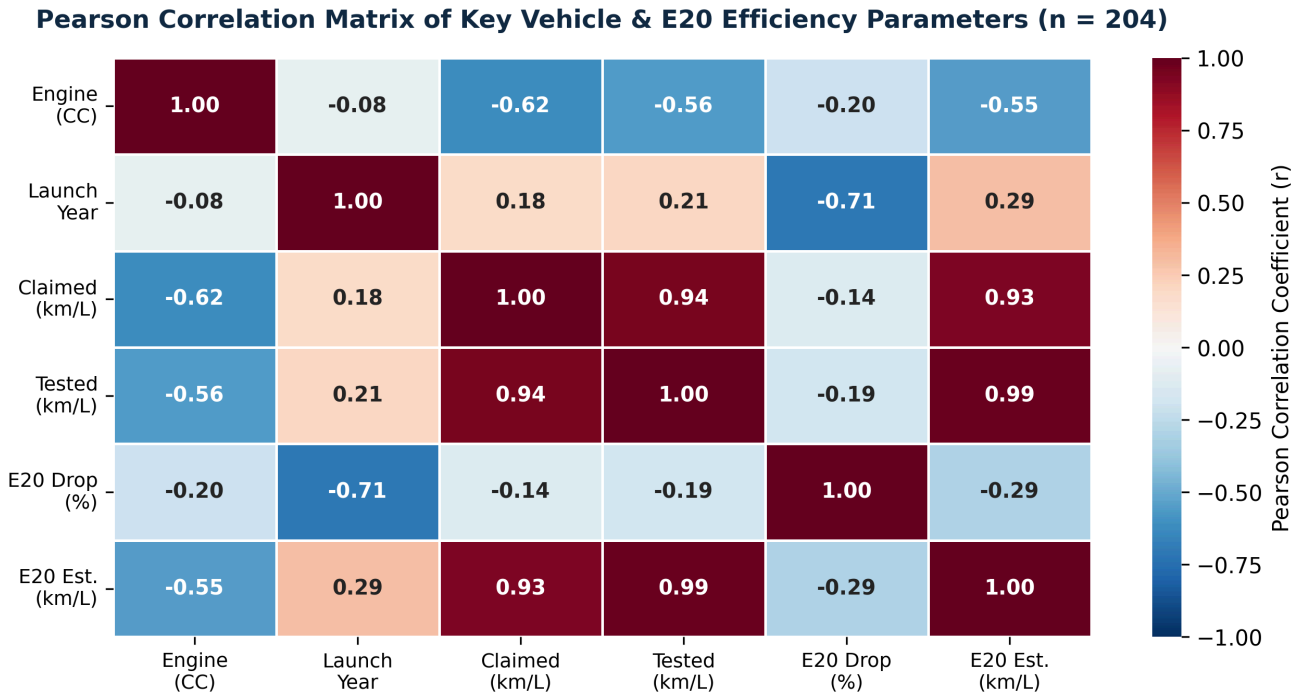


Figure 5: Annotated Pearson correlation matrix across all six continuous automotive and efficiency parameters (n = 204 validated petrol variants).

Analytical Takeaway (Figure 5): The correlation matrix synthesizes the multi-dimensional relationships within the dataset: baseline tested mileage and E20 estimated mileage move in near-lockstep (`r = +0.99`), while `Launch Year` acts as the strongest linear suppressor of `E20_Drop_%_updated` (`r = −0.71`). Meanwhile, engine displacement negatively impacts absolute mileage (`r = −0.62` claimed, `r = −0.56` tested) but weakly influences percentage E20 drop (`r = −0.20`).

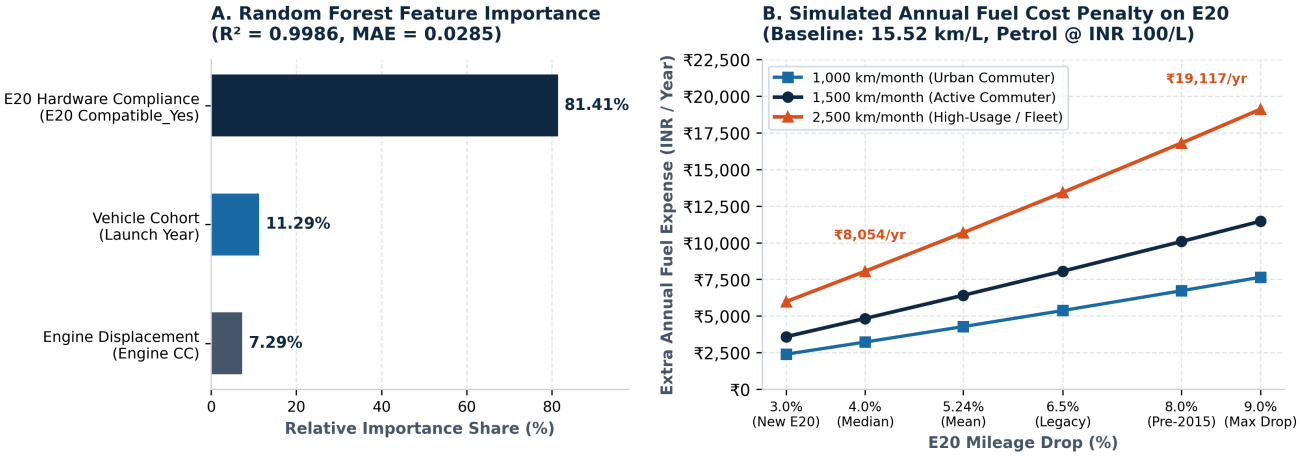


Figure 6: (A) Random Forest Regressor (R^2 = 0.9986, MAE = 0.0285) Gini importance decomposition across training predictors. (B) Simulated annual out-of-pocket fuel expense increase (INR/year) across three driving intensity tiers at a benchmark petrol price of INR 100/L and fleet mean mileage of 15.52 km/L.

Analytical Takeaway (Figure 6): In the trained `RandomForestRegressor(n_estimators=300, max_depth=8, random_state=42)` model, binary hardware compliance (`E20 Compatible_Yes`) accounts for **81.41% of total predictive importance**, followed by `Launch Year` (**11.29%**) and `Engine (CC)` (**7.29%**). Panel B translates these percentage drops into household financial impact: for an active commuter driving 1,500 km/month, a 4.0% E20 drop in a compliant car adds **₹4,832/year** (₹403/month), whereas an 8.0%–9.0% drop in a legacy pre-2015 vehicle adds **₹10,093 to ₹11,480/year** (₹841–

₹957/month). For high-utilization commercial or ride-hailing vehicles covering 2,500 km/month, a 9.0% E20 penalty inflates annual fuel expenditure by **₹19,133/year**.

---

7. Key Business & Consumer Analytics Insights (Task 7)

Synthesizing the statistical findings from **EDA.ipynb** and the deployment telemetry of **e20predictor.streamlit.app** yields six core business insights for automotive stakeholders, fleet managers, and retail consumers:

1. **Binary Hardware Compliance Dominates Efficiency Retention (81.41% Weight):** E20 compatibility is not a marketing label—it accounts for 81.41% of the variance in fuel efficiency degradation. Upgraded ECU spark-advance tables and ethanol-compatible fuel injectors restrict mileage loss to 3.0%–4.0%, cutting the financial penalty in half compared to uncalibrated engines (5.0%–9.0% loss).
2. **Generational Vulnerability of the Pre-2020 Car Parc (r = −0.71):** Because 38.5% of cataloged vehicle variants are non-E20-compatible—concentrated heavily in models launched between 2008 and 2019—legacy car owners bear a disproportionate regressive cost burden during the mandatory E20 pump transition.
3. **Persistent 18.1% Certification-to-Road Deficit:** Across 208 tested models, real-world fuel economy (15.52 km/L) trails ARAI laboratory claims (18.96 km/L) by 3.44 km/L. Evaluating E20 fuel costs using brochure mileage understates actual monthly consumer fuel expenditure by **22.2%** before E20 blending is even factored in.
4. **Sub-4-Meter Compact Concentration Amplifies Volume Impact:** With over half of Indian passenger car variants clustered around 1,197–1,199 CC (median 1,199 CC) and Maruti Suzuki plus Hyundai representing 44.3% of all variants, mass-market price-sensitive commuters represent the largest affected demographic cohort.
5. **Engine Displacement Drives Base Consumption, Not Percentage E20 Drop:** While larger engines consume substantially more fuel per kilometer ($r = -0.56$ with tested mileage), percentage E20 drop is nearly invariant to displacement ($r = -0.20$). However, because large SUVs (2,000–3,956 CC) start from a low baseline mileage (5.0–11.0 km/L), even a 4.0% E20 drop translates into a higher absolute liter penalty per 1,000 km than a compact hatchback experiencing a 6.0% drop.
6. **Ensemble Tree Regression Captures Stepwise Regulatory Regimes ($R^2 = 0.9986$):** The Random Forest architecture achieves an out-of-sample R^2 of 0.9986 and MAE of 0.0285 percentage points because decision trees naturally partition the non-linear regulatory threshold between pre-BS6, BS6 Phase-1, and BS6 Phase-2 (E20-ready) vehicle cohorts.

---

8. Data-Driven Business & Policy Recommendations (Task 7)

Based on the empirical evidence and cost simulations established in this capstone study, four targeted, quantifiable recommendations are proposed across key ecosystem stakeholders:

Target Stakeholder	Strategic Recommendation	Empirical Basis & Quantified Expected Impact
1. Automotive OEMs & Dealership Networks	Deploy Authorized E20 Retrofit Kits & ECU Calibration Updates for 2015–2022 Models	Non-compliant vehicles suffer 6.0%–9.0% E20 drops vs. 3.0%–4.0% for calibrated engines. Offering affordable elastomer fuel-line/seal kits and ECU remapping for popular 1.2L/1.5L models can recover **2.0%–3.5% in fuel efficiency**, saving active commuters ₹3,000–₹5,500 annually while boosting dealership service retention.
2. Commercial Fleet & Ride-Hailing Operators	Mandate Post-2023 E20-Compliant Procurement & Recalibrate Per-Kilometer Driver Payouts	At 2,500 km/month, a pre-2020 non-compliant sedan/hatchback incurs an extra **₹12,300–₹19,133/year** in E20 fuel costs compared to ₹6,200–₹8,050/year for an E20-compliant vehicle. Accelerating fleet replacement for pre-2018 petrol cars yields an immediate fuel OPEX reduction of **₹6,000–₹11,000 per vehicle annually**.
3. Pre-Owned Car Marketplaces & Auto Financers	Integrate E20 Compatibility Badges & Total Cost of Ownership (TCO) Calculators into Listings	Used-car buyers rarely account for the $r = -0.71$ launch-year penalty. Embedding the **E20 Expense Predictor API** directly into used-car listing portals allows buyers to compare 5-year fuel outlays alongside EMI payments, improving pricing accuracy between 2018 (non-E20) and 2023 (E20) inventory.
4. Fuel Retailers & Regulatory Bodies	Introduce Volumetric Energy-Adjusted Pricing or Pump-Level E10/E20 Consumer Labelling	Because E20 reduces fleet mean road mileage by 4.32% (from 15.52 km/L to 14.85 km/L) while retailing at identical per-liter pump prices as regular petrol, consumers face an implicit price hike per kilometer driven. Passing a **4%–5% excise-linked retail price differential** on E20 blends would neutralize the consumer mileage penalty and accelerate voluntary adoption.

9. Conclusion & Future Roadmap (Task 8)

The **E20-Expense-Predictor** capstone project demonstrates how domain-grounded data cleaning, exploratory statistical analysis, and ensemble machine learning can transform fragmented automotive testing records into a practical financial tool for everyday consumers. By curating 221 vehicle variants from ARAI, SIAM, and Autocar India, isolating 204 validated petrol powertrains, and training a high-precision Random Forest Regressor ($R^2 = 0.9986$, $MAE = 0.0285$), the project proves that vehicle E20 hardware compliance (81.41% feature weight) and launch year cohort (11.29% weight) govern fuel efficiency retention under India's 20% ethanol mandate.

Deployed live at **https://e20predictor.streamlit.app/** and documented across 21 version-controlled commits on GitHub, the application empowers Indian motorists to move beyond idealized brochure claims, quantify their exact real-world mileage degradation, and plan monthly household or fleet fuel budgets with empirical confidence. Future iterations of the pipeline will incorporate dynamic state-wise petrol price APIs, city-versus-highway driving cycle sliders, and long-term maintenance cost projections for non-E20 elastomer wear.

References & Grounded Source Links

- **Project GitHub Repository (Source Code, Notebooks & Dataset):** https://github.com/UtsavUjjwal/E20-Expense-Predictor
- **Live Deployed Streamlit Web Application:** https://e20predictor.streamlit.app/
- **Exploratory Data Analysis & ML Pipeline Notebook (`EDA.ipynb`):** https://github.com/UtsavUjjwal/E20-Expense-Predictor/blob/main/EDA.ipynb
- **Cleaned Master Dataset (`E20_predictions_updated.xlsx`):** https://github.com/UtsavUjjwal/E20-Expense-Predictor/blob/main/E20_predictions_updated.xlsx
- **Vehicle Specification Lookup Table (`car_lookup.csv`):** https://github.com/UtsavUjjwal/E20-Expense-Predictor/blob/main/car_lookup.csv
- **Streamlit Application Entry Point (`app.py`):** https://github.com/UtsavUjjwal/E20-Expense-Predictor/blob/main/app.py